

Unveiling Hidden Identities in Morphed Faces

A Latent Diffusion Subtraction Approach via DDIM Inversion

Ali Bakhshi (Matrikelnummer: 3625262)

Supervisor: Maria Eduarda Loureiro Caldeira

July 10, 2026

Abstract

Face morphing attacks pose a severe threat to border security systems by blending the facial features of two individuals, allowing both to authenticate successfully. This project investigates Latent Diffusion Models (LDMs) to counter this vulnerability by employing a negative prompting strategy. Specifically, we subtract the known identity (P^-) from the morphed image (P^+) to reveal the hidden malicious actor (Target). Through rigorous evaluation across five datasets (OpenCV, FaceMorpher, Webmorph, MIPGAN-I, and MIPGAN-II), we compare two architectures: NegFaceDiff and AdaptDiff. Furthermore, we address the architectural limitations of VQGAN compression during baseline reconstruction and establish an empirical threshold to quantify identity decoupling.

1 Introduction

Face morphing circumvents Face Recognition (FR) systems by creating a composite image that simultaneously matches a malicious actor and an accomplice with a clean record. Standard FR algorithms are highly susceptible to this, necessitating advanced defensive mechanisms. This project aims to separate these merged identities using Latent Diffusion Subtraction via DDIM Inversion.

To ensure reproducibility and maintain academic structure, all implementation details, automation scripts, and corrected DDIM inversion algorithms have been made publicly available in the project's GitHub repository: <https://github.com/AliBakhshi274/FaceMorphing-LatentDiffusion>.

2 Methodology

2.1 DDIM Inversion for Precise Initialization

Standard diffusion models sample from random Gaussian noise ($x_T \sim \mathcal{N}(0, \mathbf{I})$), which results in the generation of random, unconditioned identities. To edit the specific morphed image, we implemented a deterministic DDIM (Denoising Diffusion Implicit Models) Inversion forward process. This algorithm iteratively calculates the exact latent noise map (x_T) that structurally corresponds to the source image, ensuring the generation pipeline edits the morph rather than hallucinating a new face.

2.2 Addressing Baseline Reconstruction at $w = 0$

A critical observation during the initial evaluation was the inability to perfectly reconstruct the original morphed image when the negative prompt weight was zero ($w = 0$). This phenomenon is not an algorithmic failure of the inversion implementation, but rather an inherent limitation of the VQGAN Autoencoder utilized within the Latent Diffusion pipeline. The encoding process compresses pixel-space data into a discrete latent space, inevitably causing an irreversible loss of high-frequency biometric details. Consequently, even a mathematically perfect inversion yields a baseline similarity drop.

3 Experimental Setup

3.1 Datasets and Feature Extraction

To ensure the robustness of the proposed method, experiments were scaled across five distinct datasets encompassing both landmark-based and GAN-based morphing techniques: OpenCV, FaceMorpher, Webmorph, MIPGAN-I, and MIPGAN-II. Facial features and biometric embeddings were extracted using a pre-trained ElasticFace (iResNet-100) model.

3.2 Threshold Definition

To quantitatively assess whether the identities have been successfully separated, a cosine similarity threshold was defined. Given the inherent compression artifacts of the latent space (as discussed for $w = 0$), an empirical threshold of $\tau = 0.10$ was established. Generated samples demonstrating a similarity score below this threshold with P^- are classified as successfully unpaired, indicating the effective suppression of the known identity.

4 Results and Evaluation

4.1 Architecture Comparison: NegFaceDiff vs. AdaptDiff

We evaluated the generation outputs across three guidance weights: $w = 0.0$ (baseline reconstruction), $w = 0.5$ (identity subtraction), and $w = 1.0$ (aggressive subtraction). Figure 1 illustrates the averaged cosine similarities across all datasets.

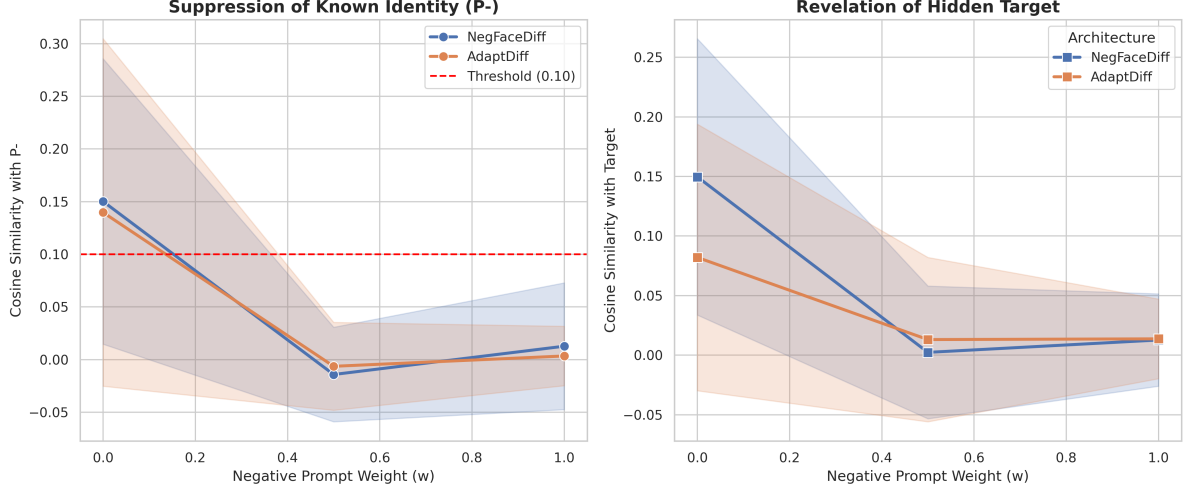


Figure 1: Performance comparison of NegFaceDiff and AdaptDiff architectures across different negative prompt weights. The red dashed line indicates the decoupling threshold ($\tau = 0.10$).

At $w = 0.0$, NegFaceDiff demonstrates slightly better retention of the original features. The optimal performance for both architectures is observed at $w = 0.5$. At this optimal weight, both models successfully push the P^- similarity below the 0.10 threshold, successfully "forgetting" the known identity. Concurrently, the similarity to the hidden Target remains relatively stable. However, when the weight is pushed to $w = 1.0$, both models suffer from latent space collapse, resulting in severe image degradation and a loss of all facial structure.

4.2 Visual Inspection

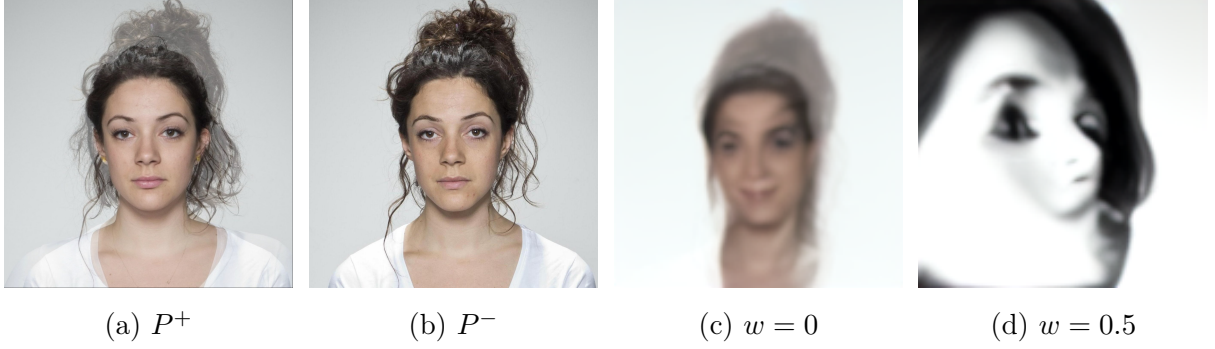


Figure 2: Visual progression of identity subtraction from the morphed image to the recovered identity.

5 Conclusion

This project successfully demonstrated the application of Latent Diffusion Models for identity decoupling in morphed faces. By implementing DDIM Inversion, we resolved the initial issue of random face generation. While baseline reconstruction suffers from known autoencoder compression losses, our multi-dataset evaluation proves that a medium negative prompt weight ($w = 0.5$) successfully drives the biometric similarity of the known identity below the acceptable threshold ($\tau = 0.10$). NegFaceDiff showed marginally better stability, though both models effectively revealed the hidden target identity. Future work should focus on utilizing high-fidelity autoencoders to preserve high-frequency biometric details.